

SKELLEFTEA, Sweden

WMO: 022930

Lat: 64.624N

Lon: 21.072E

Elev: 48

StdP: 100.75

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.3	(c) -21.8	(d) -28.1	(e) 0.3	(f) -25.4	(g) -24.0	(h) 0.4	(i) -21.6	(j) 9.9	(k) 1.7	(l) 8.4	(m) -0.6	(n) 1.6	(o) 260	(p) 0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	26.3	17.0	24.2	16.2	22.1	15.5	18.7	23.3	17.6	22.1	16.5	20.6	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.3	20.4	15.9	11.4	19.2	14.9	10.7	18.2	53.4	23.4	49.7	22.2	46.5	20.6	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.4	6.5		-28.9	28.4	4.4	2.1	-32.1	29.9	-34.6	31.2	-37.1	32.4	-40.3	33.9
			DB	-29.1	20.1	4.4	1.5	-32.2	21.2	-34.8	22.1	-37.2	22.9	-40.4	24.0
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-8.0	-7.8	-3.7	2.1	8.1	13.0	16.3	14.6	10.0	3.5	-1.5	-4.5
	DBStd	9.62	6.54	7.17	5.55	3.50	4.20	3.55	3.03	3.12	3.30	3.92	4.87	5.72
	HDD10.0	2855	559	498	423	237	86	11	0	2	41	203	345	450
	HDD18.3	5409	817	731	682	486	316	164	79	121	251	460	595	708
	CDD10.0	513	0	0	0	1	28	101	195	146	41	1	0	0
	CDD18.3	26	0	0	0	0	0	4	16	6	0	0	0	0
	CDH23.3	279	0	0	0	0	13	57	153	56	1	0	0	0
	CDH26.7	41	0	0	0	0	1	10	24	7	0	0	0	0
Wind	WSAvg	3.0	2.8	2.9	3.2	3.0	3.2	3.2	2.8	2.7	2.9	3.0	2.9	3.0
Precipitation	PrecAvg	596	45	33	33	34	39	51	59	71	60	59	62	48
	PrecMax	840	94	62	63	81	97	93	144	130	123	164	117	103
	PrecMin	453	5	5	9	8	12	16	5	15	11	9	19	6
	PrecStd	98	21	18	15	19	22	21	31	33	28	34	29	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	7.0	10.1	17.0	25.1	28.1	29.0	27.9	21.1	13.8	9.8	6.9
		MCWB	3.3	4.1	5.3	10.0	14.5	16.6	18.7	19.3	14.6	10.5	7.9	4.5
	2%	DB	3.1	5.0	7.5	13.2	21.9	24.9	27.0	25.0	18.9	12.0	7.8	5.1
		MCWB	1.7	2.6	3.8	7.1	13.1	15.0	17.9	16.7	13.9	9.2	6.6	3.5
	5%	DB	1.9	2.8	5.8	10.8	18.6	21.9	25.0	22.7	17.0	10.2	6.0	3.9
		MCWB	1.2	1.1	2.7	5.5	11.4	14.2	16.8	16.1	12.8	8.4	4.8	2.8
	10%	DB	0.8	1.1	4.0	8.6	16.0	19.8	22.8	20.6	15.3	9.1	4.8	2.5
		MCWB	0.3	0.4	1.4	4.2	10.1	12.9	16.5	15.7	12.2	7.6	3.9	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	4.4	5.3	10.1	15.4	18.2	20.9	20.2	15.9	11.5	7.9	4.7
		MCDB	4.8	6.7	9.2	16.2	22.3	24.9	25.4	25.5	18.5	13.0	8.8	5.6
	2%	WB	1.8	2.8	4.0	7.4	13.6	16.3	19.5	18.3	14.7	9.9	6.5	3.7
		MCDB	2.4	4.5	7.2	12.3	20.5	22.1	23.5	22.4	17.5	11.1	7.3	4.7
	5%	WB	1.0	1.4	2.8	5.8	12.0	15.1	18.2	17.2	13.7	8.8	4.9	2.7
		MCDB	1.5	2.3	5.2	9.9	17.9	19.8	22.8	20.9	16.0	10.0	5.4	3.5
	10%	WB	0.1	0.3	1.6	4.5	10.5	13.9	17.1	16.2	12.9	7.7	3.7	1.6
		MCDB	0.6	0.9	3.3	7.8	15.2	18.2	21.6	19.5	14.8	8.9	4.2	2.3
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.9	9.1	9.1	9.8	9.1	9.0	8.8	8.0	6.0	5.2	6.3
		MCDBR	6.6	7.4	9.8	13.4	14.8	12.8	12.7	12.3	10.0	7.4	5.1	5.9
		MCWBR	6.0	6.1	6.8	8.1	7.4	5.9	5.4	5.7	6.0	5.4	4.4	5.4
	5% WB	MCDBR	6.2	6.4	8.9	12.4	14.0	10.8	10.8	9.8	8.4	6.5	4.8	5.6
		MCWBR	5.8	5.5	6.3	7.7	7.3	5.7	5.4	5.3	5.6	5.2	4.5	5.2
Clear-Sky Solar Irradiance	taub	N/A	0.276	0.304	0.337	0.343	0.345	0.361	0.349	0.320	0.302	0.339	0.348	
	taud	N/A	2.166	2.233	2.308	2.459	2.496	2.473	2.500	2.558	2.401	2.607	2.633	
	Ebn at Noon	N/A	621	774	834	868	872	844	821	777	605	310	147	
	Edh at Noon	N/A	58	84	99	94	93	93	83	63	47	22	13	
All-Sky Solar Radiation	RadAvg	0.12	0.68	2.15	3.79	4.95	5.50	5.06	3.81	2.23	0.86	0.19	0.04	
	RadStd	0.01	0.12	0.22	0.33	0.50	0.44	0.47	0.35	0.29	0.15	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page